MICROWAVE PLASMA GUIDE FOR LAB-GROWN DIAMOND APPLICATIONS

Together, we reinvent your processes with our
Microwave and Radio Frequency solutions since 1978

microwave & radio frequency

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MICROWAVE PLASMA

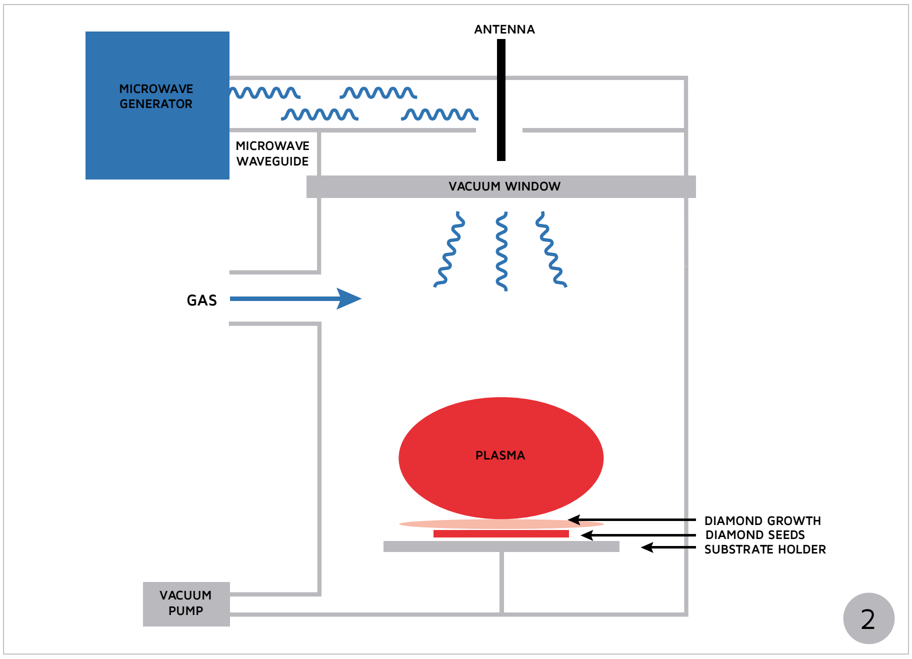
To generate a plasma, microwave energy oscillates the electrons present in the gas (mostly H_2 and C_2H_2 or CH_4 in the case of diamonds) which in turn produces ions by colliding with other atoms and molecules in the surrounding. Typically, 2450 MHz frequency is used as the excitation source but 915 MHz generators are becoming more and more popular due to their higher power.

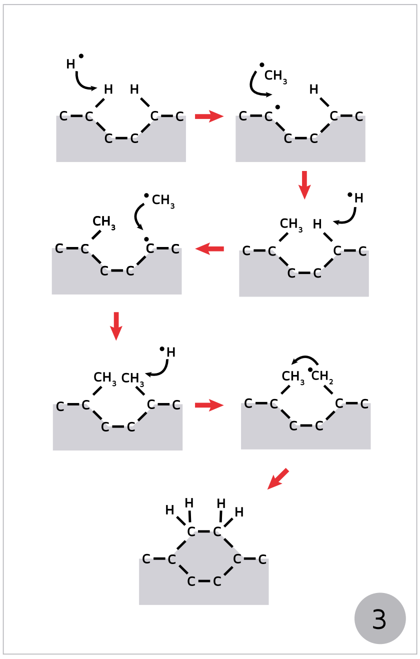
For MPCVD, the plasma is created in a resonant cavity, allowing to generate a plasma ball (1) in levitation just above the substrate holder, where the diamond structure will grow.

The microwave plasma is used to activate the hydrocarbon feed and dissociate molecular hydrogen. The size of the plasma ball increases as more the microwave power is applied. Intimate contact of the plasma ball with the substrate is not essential for diamond deposition by MPCVD. Microwave Plasma Chemical Vapor Deposition is the most efficient method for synthetic diamond production.

SYNTHESIS OF LAB-GROWN DIAMOND

Synthetic diamonds are generated by a Plasma Ehanced Chemical Vapor Deposition (PECVD) process in a microwave resonant reactor (2) :



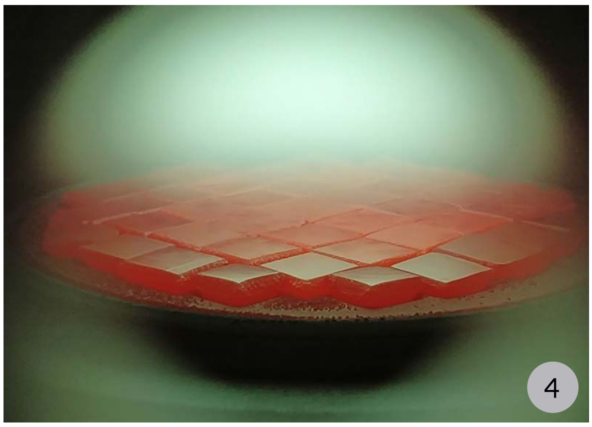


The plasma, created by the ionisation of H_2 / C_2H_2 or H_2 / CH_4 gas by microwaves, generate a carbon deposition due to the creation of reactive species in the microwave plasma. The activated carbon-hydrogen species attaches itself to the seed atom (3). This process repeats itself endlessly for 21 to 28 days to replicate the crystal structure of the diamond seed crystal in three dimensions. All the process is happening under reduced pressure and a temperature between 900 to 1200 °C for the diamonds to grow properly (5).

The keys to perfection

Diamond synthesis is a process that requires a very high quality and reliable generator as it last several weeks. The generators must run round the clock without a glitch, and **whistand important power variations** as any plasma unstability may create imperfections in the diamonds.

In order to guarantee their reliability, microwave generators dedicated to the diamond industry **must be 100 % digital** and with numerous sensors and warnings to prevent microwave power loss. They must also collect data on their performances.



Finally, the **quality of the magnetron** is essential to offer a stable microwave spectrum to produce the purest lab-grown diamonds possible and perfect its management.

SAIREM'S SOLUTIONS FOR CVD

As the microwave plasma expert, SAIREM provides a range of microwave generators with different power supplies, stub tuners, waveguides and accessories for the synthetic diamond industry.

Microwave generators

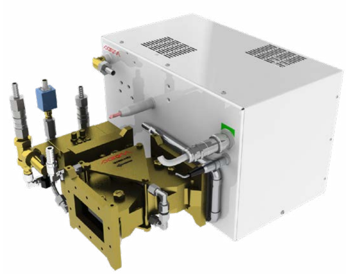
Our industrial microwave generators are designed for processes that can last several weeks. They provide a very stable microwave spectrum which is mandatory for the generation of high quality diamonds. They offer among the **best power efficiency** of the market (85%), low ripple and extended magnetron lifespan.
Besides, all our generators are equipped with preventive alarm management and an **Automatic Restart Function (ARF)** which preserves the plasma integrity in case of electrical micro-cuts. To guarantee the best stability possible, SAIREM's generators are 100 % digital and equiped with **user friendly touch interface.** They also collect process data, so our customers can analyse their cycle performances.

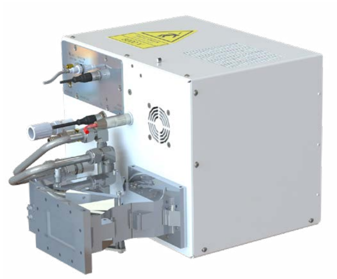
915 MHz microwave generators

	GLP360	GLP540
Power	36 kW	54 kW
Power type	Switch mode	Switch mode
Power efficiency	85 %	85 %
Power variation	0.1 kW [1]	0.1 kW [1]
Frequency	915 MHz	915 MHz
Capacity	Up to 100 seeds [2]	Up to 150 seeds [2]
Automatic Restart Function (ARF)	Yes	Yes
Mains	400 V - 50 / 60 Hz	400 V - 50 / 60 Hz
Voltage variation tolerance	±10 %	±10 %

[1] From 10% to 100%   [2] For 1 carat polished diamond

2450 MHz microwave generators





	GMP60	GMP100
Power	6 kW	10 kW
Power type	Switch mode	Switch mode
Power efficiency	85 %	85 %
Power variation	0.01 kW [1]	0.01 kW [1]
Frequency	2450 MHz	2450 MHz
Capacity	Up to 20 seeds [2]	Up to 35 seeds [2]
Automatic Restart Function (ARF)	Yes	Yes
Mains	208 / 400 / 480 V - 50 / 60 Hz	400 / 480 V - 50 / 60 Hz
Voltage variation tolerance	±10 %	±10 %

Services

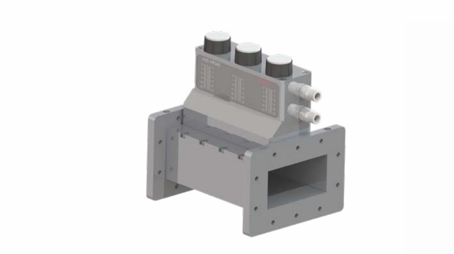
- Technological consulting.
- Reactors simulations and modelizations.
- Comprehensive commissioning and after-sales service.

[1] From 10% to 100%

[2] For 1 carat polished diamond

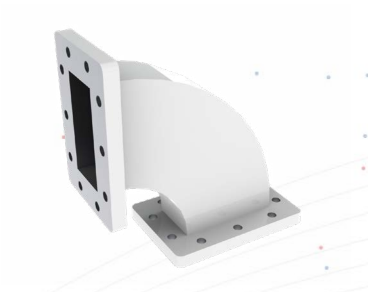
Impedance stub tuners

Our compact impedance tuners can be manual or fully automatic. They are both designed to decrease the level of reflected power in a microwave installation. They are adaptable to both 915 MHz and 2 450 MHz and equipped with a water cooling system.



	AI3S	AI4S	AI4S
Maximum Power	10 kW	10 kW	100 kW
Frequency	2450 MHz	2450 MHz	915 MHz
Mode	Manual	Automatic	Automatic

Accessories



Waveguide

Different shapes are available.

Sliding short circuit

- For tuning applications.
- Optimize power transmision to plasma.

Microwave survey metter

- Detect microwave leakage.
- Automatically stop the machine.

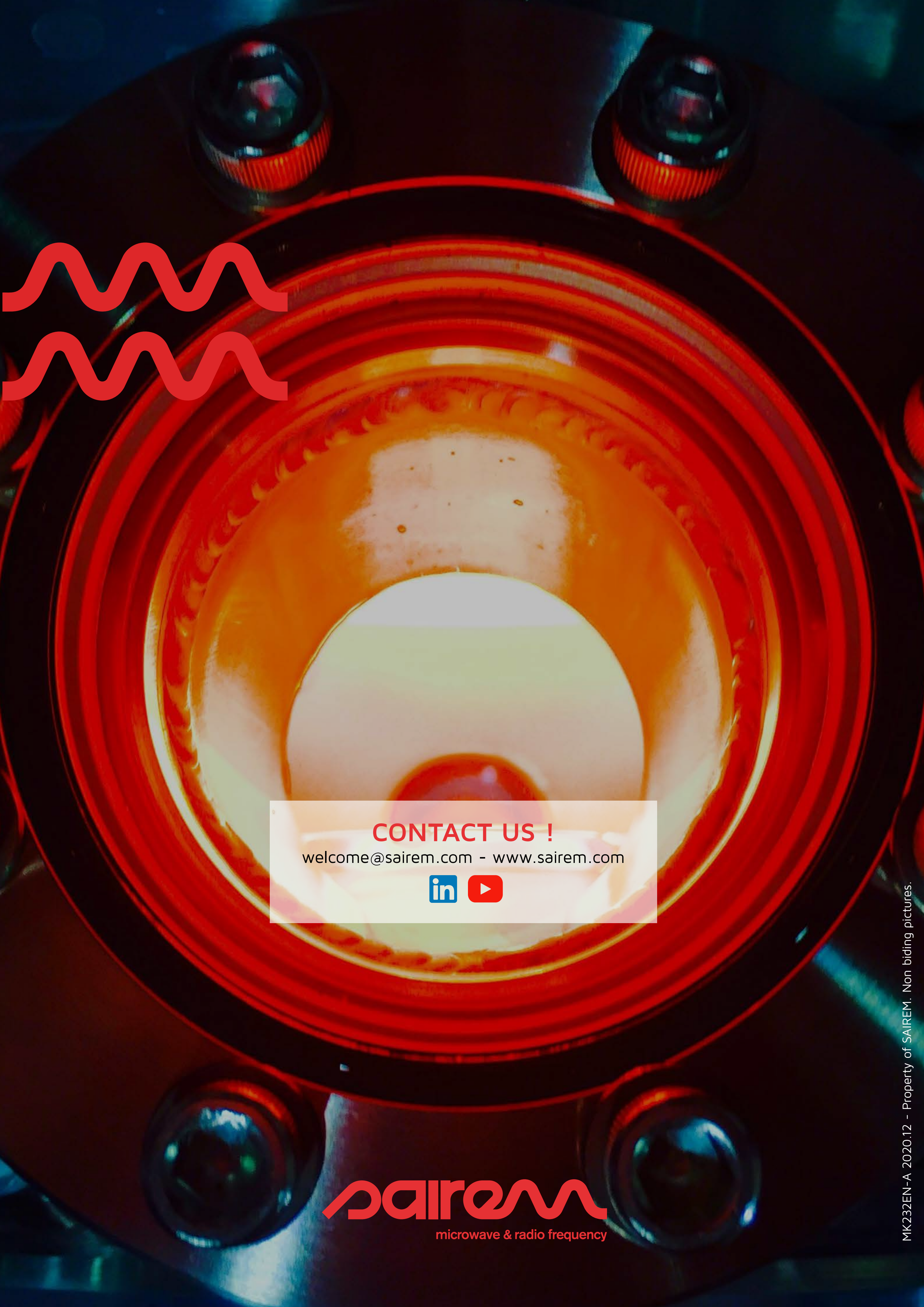
ABOUT SAIREM

SAIREM is a world leader in industrial microwave and radio-frequency applications, with more than 5000 references in operation in 70 countries, from the standalone 200 W solid-state generator to the fully automated processing line delivering 700 kW.

Today, SAIREM offers the most advanced range of industrial microwave and radio frequency equipment for thermal processing and plasma generation.

Key figures

- Created in 1978
- 70+ employees
- 5 000 machines across 70 countries
- 8 nationalities in our team
- 15 M€ yearly turnover
- 90 % export share
- 10 % invested in R&D



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